Exp. No: 12(a) Echo client - Service using TCP sockets.

AIM: To implement an echo client-server application using TCP sockets

ALGORITHM:

Server

1. Create a TCP socket
2. Bind the it to an IP addresses and port
3. Listen for client connections
4. Accept the client connection
5. Receive the message and send it back
6. close the connection

Client

1. Create a TCP socket
2. Connect to the server
3. Send a message.
4. Receive a echoed message
5. Display msg
6. Close the connection

Code:

Server

```
import socket
S = socket.socket()
S.bind("localhost", 5000)
S.listen(1)
C, addr = s.accept()
msg = C.recv(1024).decode()
C.send(msg.encode())
```

3

Input:

Enter message: Hello server

Output:

Server: Hello server

```
c.close()
s.close()
```

client

```
import socket
s = socket.socket()
s.bind("localhost", 5000)
s.listen(1)
c, addr = s.accept()
msg = c.recv(1024).decode()
c.send(msg.encode())
c.close()
s.close()
```

client

```
import socket
s = socket.socket()
s.connect(("localhost",5000
msg = input("E
```

client:

```
import socket
s = socket.socket()
s.connect(("localhost", 5000))
msg = input("Enter message:")
s.send(msg.encode())
print("Echo:", s.recv(1024).decode())
s.close()
```

Result